E-COMMERCE ANALYTICS PROJECT

Market Basket Analysis

Understanding Customer Behavior, Segments & Purchase Patterns

Video Link: https://www.loom.com/share/3050ba64daa649cd8f5183c65116c995

Python · Pandas · NumPy · Matplotlib · scikit-learn (K-Means)

Solved Project Walkthrough — Data Cleaning to Customer Segmentation

PROJECT OVERVIEW

A Five-Stage Analytics Pipeline

1

Data Cleaning

Fix column names, strip whitespace, standardize categories, handle missing values

2

Descriptive Analysis

Demographics, purchase habits, browsing behavior & satisfaction trends

3

Customer Segmentation

K-Means clustering on purchase frequency & satisfaction to build 4 profiles

4

Recommendation Insights

Link personalization & review reliability to satisfaction and rating accuracy

5

Visualization & Reporting

Consolidated charts summarizing behavior, segments and key drivers

THE DATA

Dataset & Tools

800	24	0	0
Customer Responses	Survey Columns	Duplicate Rows	Missing Values After Cleaning

Source File: Target.csv → Cleaned Output: Target_Cleaned.csv

TOOLS & LIBRARIES

pandas / numpy	Data loading, cleaning & aggregation
matplotlib	Charting descriptive & segment visuals
scikit-learn	StandardScaler + KMeans clustering

Cleaning the Raw Survey Data

```
# Fix a duplicated/misnamed column & strip header whitespace
df.columns.values[17] = "Personalized_Recommendation_Frequency_Rating"
df.columns = df.columns.str.strip()

# Trim stray whitespace from every categorical column
for column in categorical_columns:
    df[column] = df[column].str.strip()

# Standardize inconsistent category labels
df["Product_Search_Method"] = df["Product_Search_Method"] \
    .replace("categories", "Categories").fillna("Unknown")

# Coerce rating columns to numeric
for column in numeric_columns:
    df[column] = pd.to_numeric(df[column], errors="coerce")

df.to_csv("Target_Cleaned.csv", index=False)
```

WHAT THIS FIXED

- Renamed a duplicated survey column so ratings map correctly
- Removed leading/trailing spaces from headers & values
- "categories" and "Categories" merged into one label
- Missing search-method entries filled as Unknown
- Rating columns converted from text to numeric

Making charts

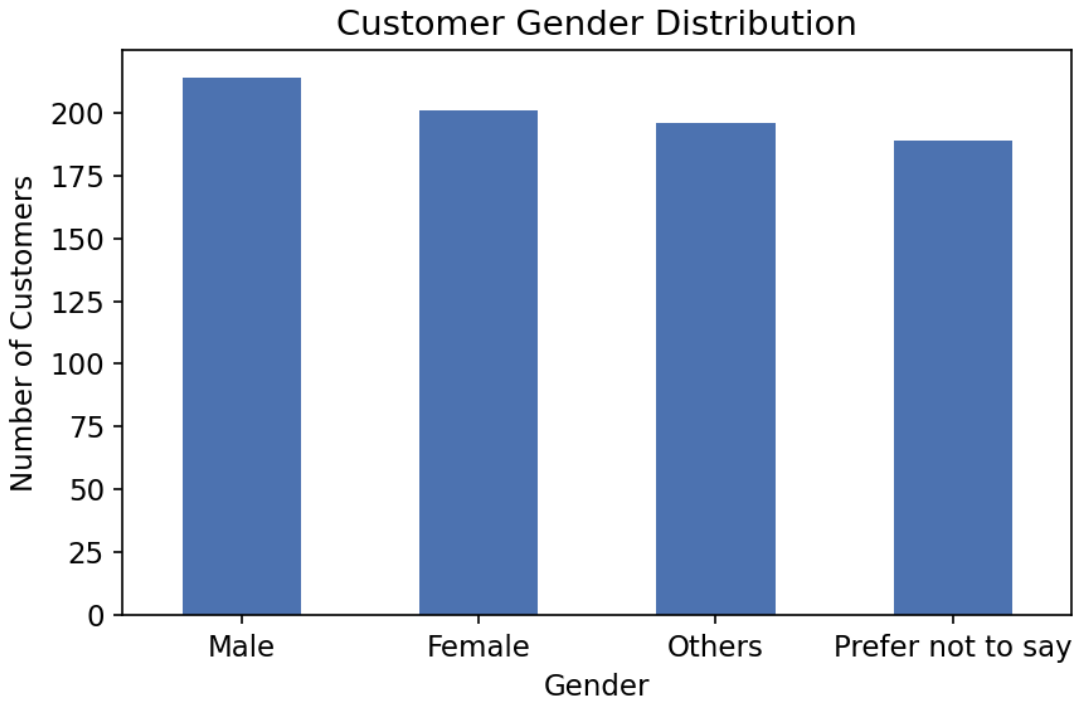
```
# Age summary
print("\n===== AGE SUMMARY =====")
print(df["age"].describe())

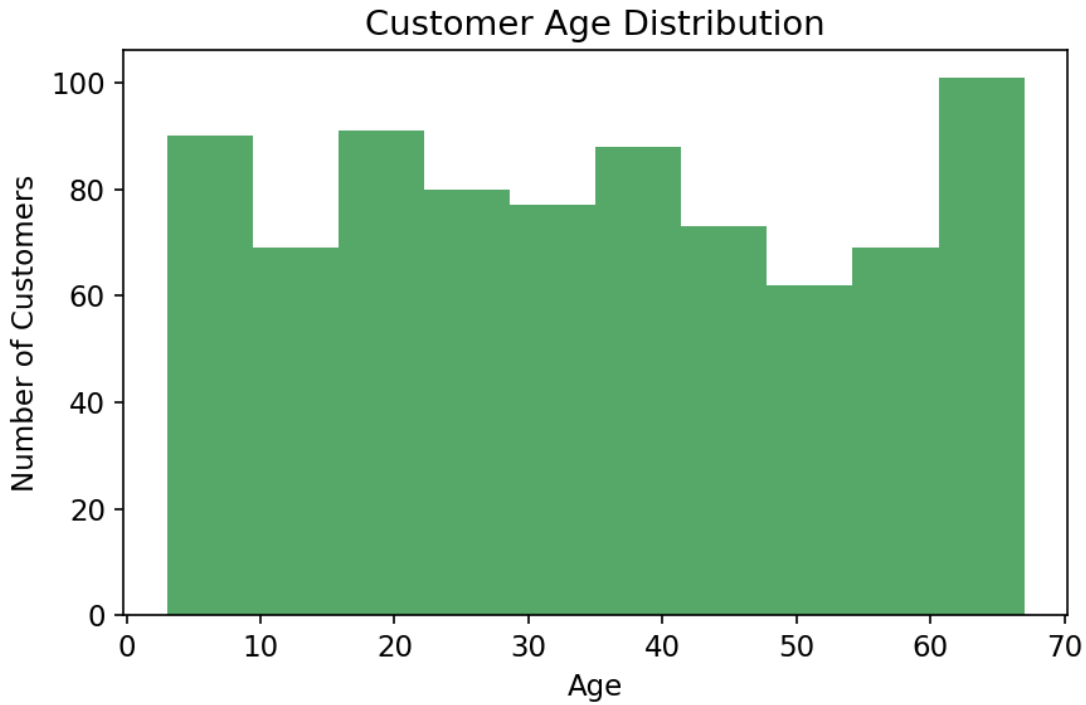

print("\nMean Age:", round(df["age"].mean(), 2))
print("Median Age:", df["age"].median())
print("Minimum Age:", df["age"].min())
print("Maximum Age:", df["age"].max())
# Gender distribution
gender_counts = df["Gender"].value_counts()


print("\n===== GENDER DISTRIBUTION =====")
print(gender_counts)


print("\nGender Percentage:")
print((gender_counts / len(df) * 100).round(2))
```

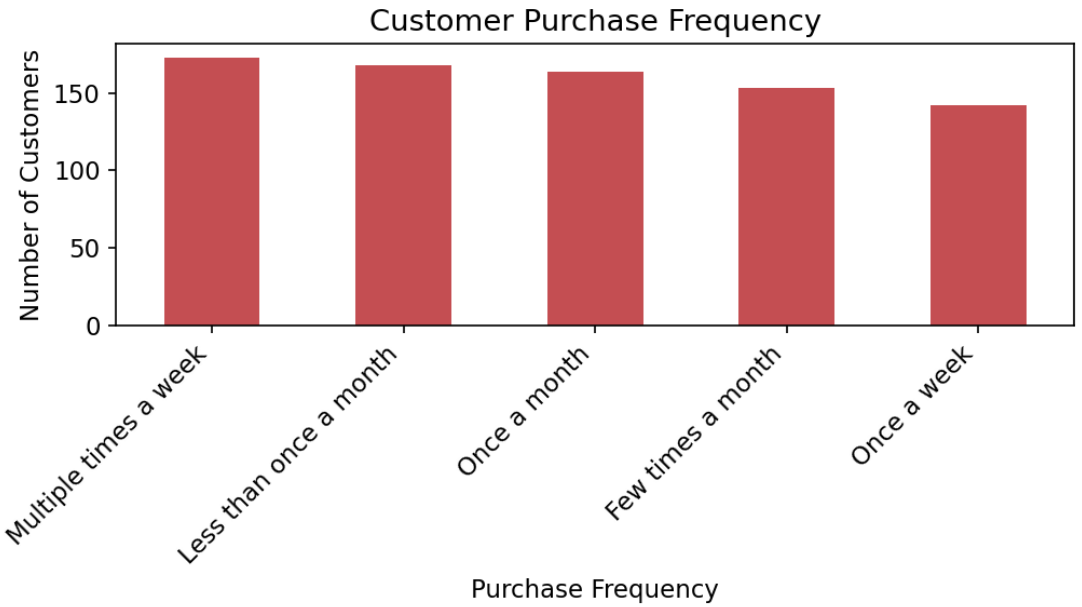
Who Are the Customers?

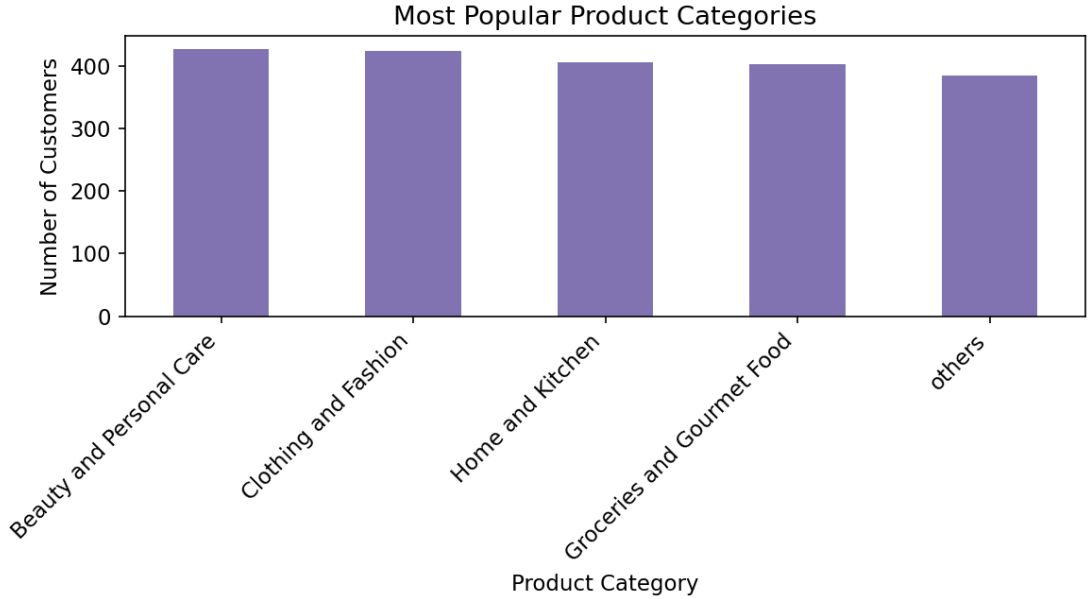




Average age 34.7 (median 34, range 3–67) · Gender split is nearly even across Male, Female, Others & Prefer not to say

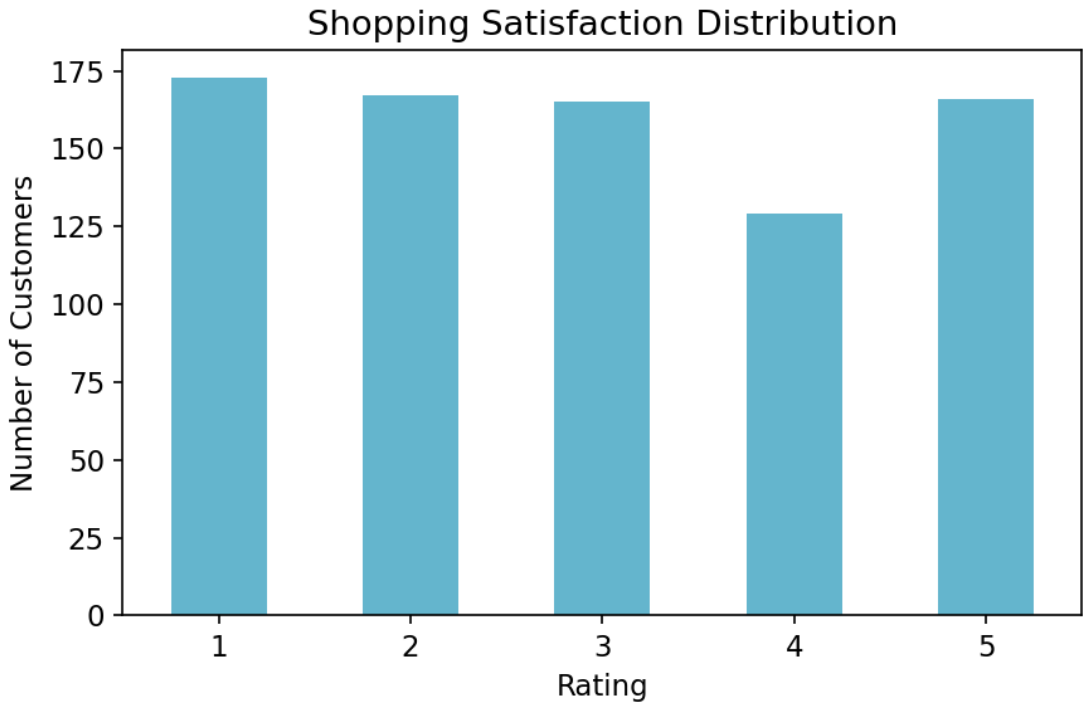
How Often — and What — Do They Buy?

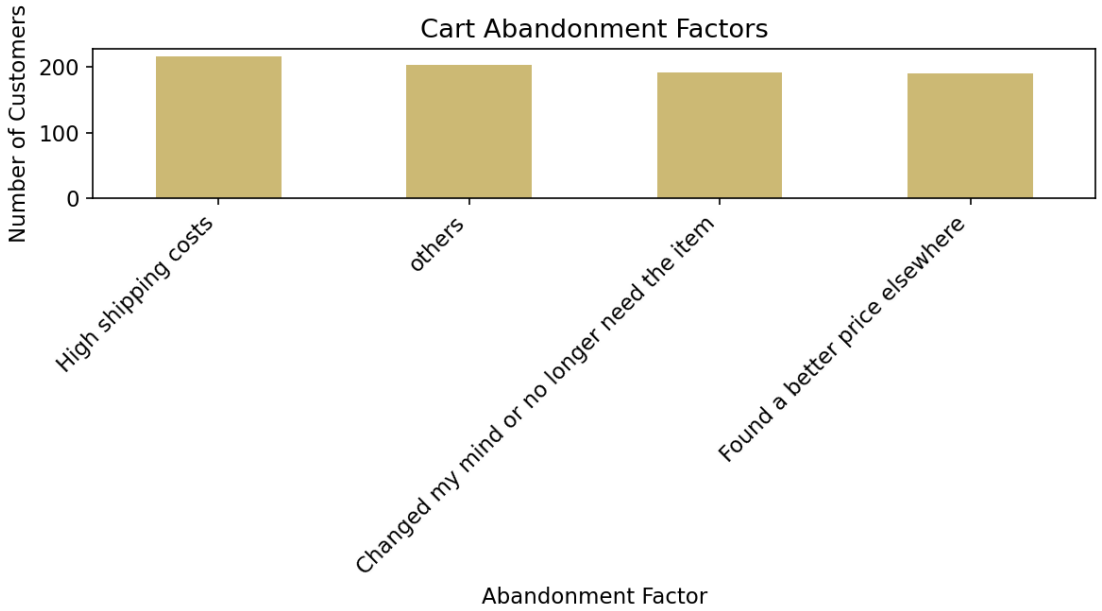




Purchase frequency is broadly spread ("Multiple times a week" leads at 173 responses) · Beauty & Personal Care and Clothing & Fashion are the top categories

Satisfaction & Cart Abandonment





Satisfaction ratings are fairly evenly distributed 1–5 · High shipping costs is the single largest cart-abandonment factor (216 responses)

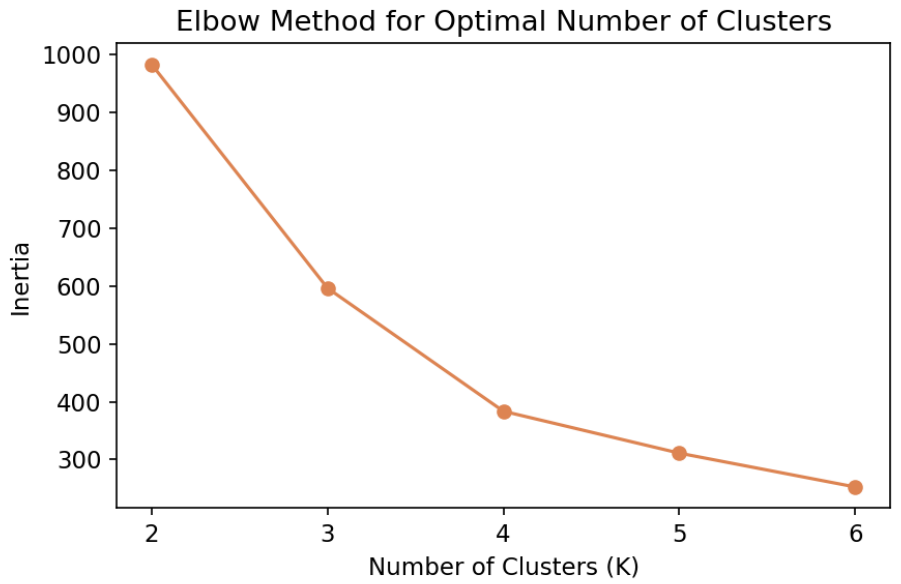
Building the K-Means Model

```
freq_map = {"Less than once a month": 1, "Once a month": 2,
            "Few times a month": 3, "Once a week": 4,
            "Multiple times a week": 5}
df["Purchase_Frequency_Score"] = df["Purchase_Frequency"].map(freq_map)

X = df[["Purchase_Frequency_Score", "Shopping_Satisfaction"]]
scaled = StandardScaler().fit_transform(X)

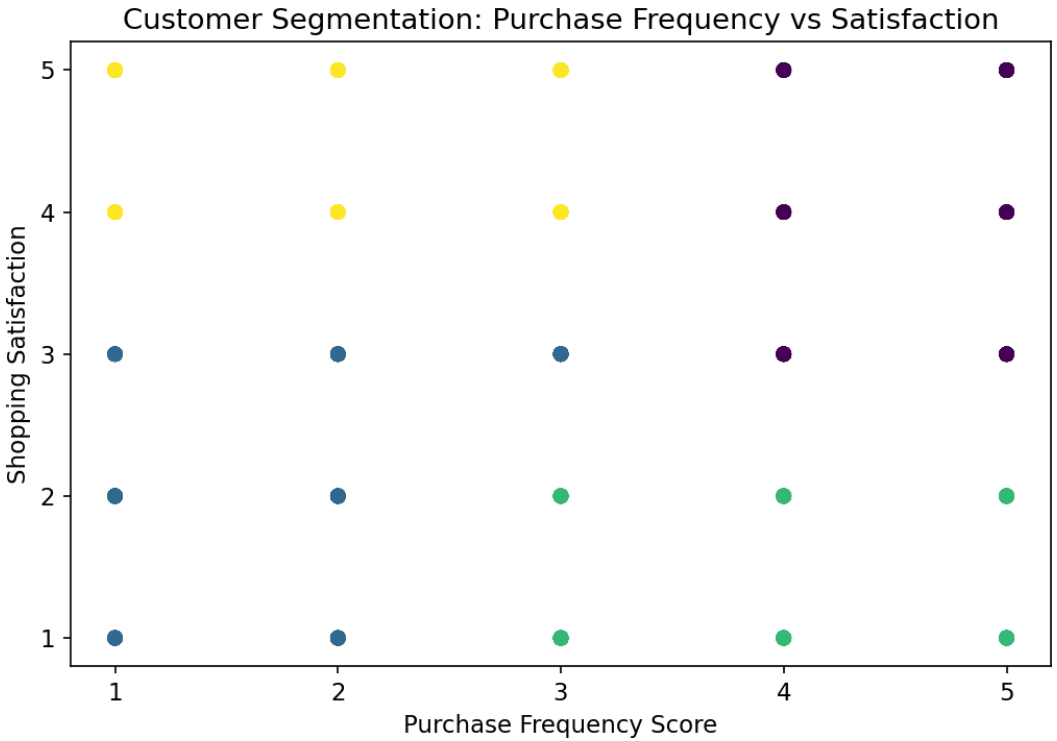
# Elbow method to choose K
inertia = [KMeans(n_clusters=k, random_state=42, n_init=10)
           .fit(scaled).inertia_ for k in range(2, 7)]

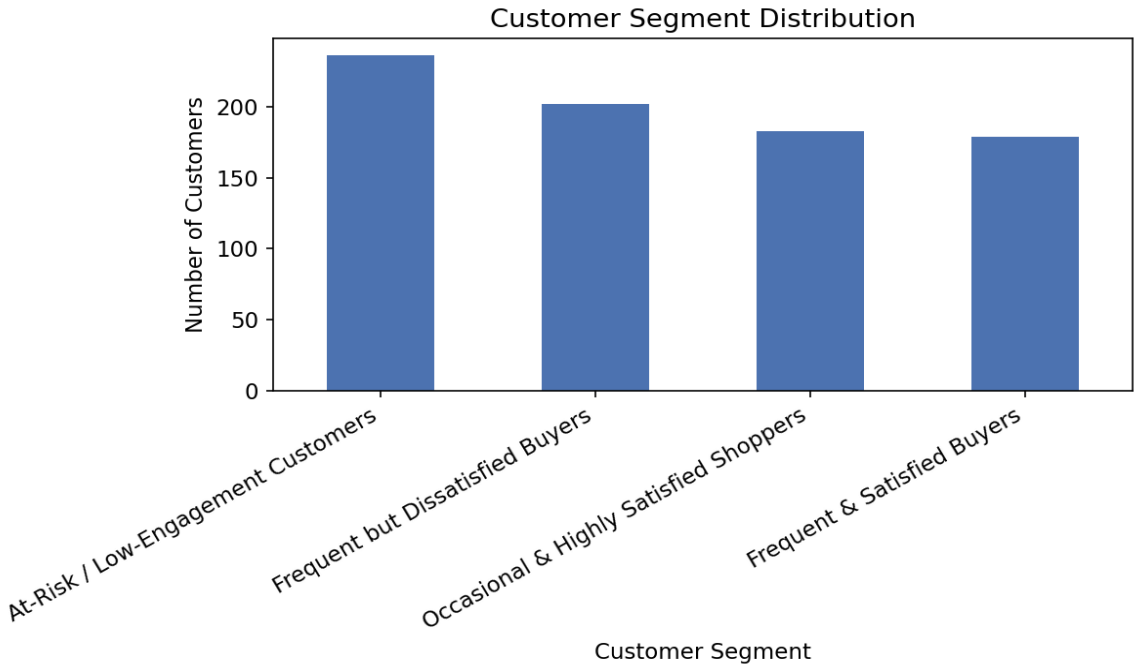
# Fit final model with K = 4
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
df["Cluster"] = kmeans.fit_predict(scaled)
```



K = 4 gives a clear bend in inertia — the chosen number of clusters.

Four Distinct Customer Segments





Clusters separate cleanly along purchase frequency and satisfaction, forming four almost-equal-sized behavioral groups (22–30% of customers each).

Segment Profiles at a Glance

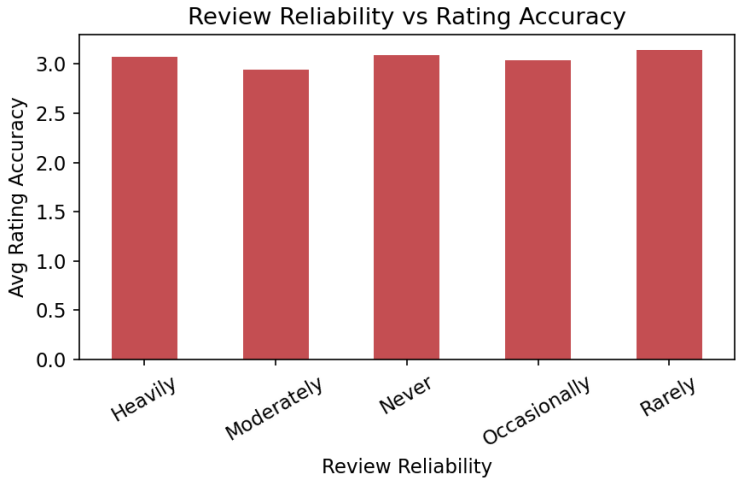
Segment	Customers	% of Total	Avg Purchase Frequency (1-5)	Avg Satisfaction (1-5)
Frequent & Satisfied Buyers	179	22.4%	4.55	3.97
Frequent but Dissatisfied Buyers	202	25.3%	4.04	1.46
Occasional & Highly Satisfied Shoppers	183	22.9%	1.97	4.57
At-Risk / Low-Engagement Customers	236	29.5%	1.68	2.14

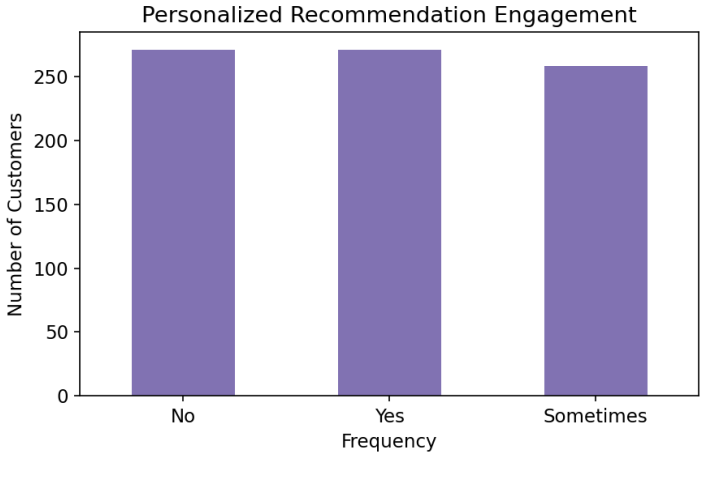
Reading the segments

- Frequent & Satisfied — the loyalty base; keep them happy and reward repeat purchases
- Frequent but Dissatisfied — highest purchase activity yet lowest satisfaction: a retention risk worth investigating
- Occasional & Highly Satisfied — infrequent buyers who love the experience; strong upsell candidates
- At-Risk / Low-Engagement — the largest group; low frequency and low satisfaction need re-engagement

Do Recommendations and Reviews Build Trust?







Customers who find recommendations more helpful report higher shopping satisfaction, and those who trust reviews more also rate product-listing accuracy higher — engagement with personalization tracks with overall trust in the platform.

SUMMARY

Key Findings

1 Clean, reliable data

800 responses, 0 duplicates and 0 missing values after standardizing labels and coercing rating columns.

2 Balanced customer base

Age, gender and purchase frequency are all broadly spread — no single dominant segment in the raw demographics.

3 Shipping cost is the #1 friction point

216 customers cite high shipping costs as their main reason for cart abandonment.

4 Four actionable segments

K-Means (K=4) splits customers by purchase frequency & satisfaction into loyal, at-risk, dissatisfied-but-active, and occasional-but-happy groups.

5 Trust drives satisfaction

Recommendation helpfulness and review reliability both correlate positively with satisfaction and rating accuracy.

Thank You

Questions & Discussion — Market Basket Analysis Project

Full code: Market_Basket_Analysis.py | Data: Target.csv → Target_Cleaned.csv